

CHIGNIK, AK, USA

WMO: 703333

Lat: 56.311N

Lon: 158.373W

Elev: 6

StdP: 101.26

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 25518

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-12.5	-11.0	-27.3	0.3	-1.6	-24.1	0.4	2.3	15.6	-8.3	14.6	-7.6	8.5	270	0.748

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.3	17.7	13.1	16.4	12.4	15.2	11.9	14.1	16.8	13.2	15.8	12.4	14.6	4.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	9.1	15.1	12.1	8.8	14.5	11.2	8.3	13.8	39.4	16.8	37.1	15.6	35.1	14.8	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.6	10.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	0.0	1.6	-1.2	2.2	6.3	9.4	11.8	12.6	10.3	6.1	2.7	-0.1
	DBStd	5.79	4.65	3.59	5.01	3.08	2.70	1.93	1.64	1.77	1.93	2.97	3.51	4.47
	HDD10.0	1955	311	234	349	233	119	32	2	1	18	123	220	312
	HDD18.3	4808	569	468	606	483	374	267	202	177	241	379	470	571
	CDD10.0	191	0	0	1	0	4	15	58	83	27	3	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.3	4.4	4.3	5.3	4.4	3.9	3.4	3.2	3.6	4.0	4.6	4.9	5.1
Precipitation	PrecAvg	909	82	69	56	48	51	61	78	104	97	88	85	90
	PrecMax	1193	278	160	117	92	100	96	135	155	150	161	143	178
	PrecMin	698	20	12	9	12	11	33	30	56	48	32	29	32
	PrecStd	130	44	31	29	18	22	17	22	22	24	30	30	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	7.5	13.9	9.9	17.7	20.2	19.1	19.8	17.0	13.0	11.1	8.0
		MCWB	4.1	4.8	7.7	5.6	10.4	13.8	13.5	15.7	12.7	10.2	8.5	5.5
	2%	DB	6.3	6.4	7.6	7.9	13.7	16.2	17.3	17.8	15.0	12.2	8.9	7.3
		MCWB	4.2	4.0	4.4	4.7	8.5	10.8	12.9	13.9	11.7	9.7	7.1	5.1
	5%	DB	6.0	6.0	6.1	7.2	11.4	13.8	16.0	16.9	13.8	11.2	7.8	6.2
		MCWB	4.0	3.8	3.9	4.2	7.5	9.7	12.3	13.3	10.9	9.0	6.0	4.1
	10%	DB	5.0	5.1	4.8	6.1	9.5	12.5	14.1	15.8	12.9	10.1	7.3	5.1
		MCWB	3.4	3.2	3.0	3.5	6.8	9.2	11.1	12.7	10.2	8.1	5.2	3.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	5.7	8.0	6.9	10.7	13.9	14.7	16.4	13.7	11.2	8.5	6.8
		MCDB	6.8	7.3	14.0	8.4	16.5	19.4	18.0	19.3	16.0	12.5	9.6	7.5
	2%	WB	4.9	5.0	5.6	5.6	9.1	11.5	13.5	14.6	12.3	10.2	7.6	5.4
		MCDB	6.2	6.4	7.5	7.3	13.3	15.1	16.4	17.1	14.2	11.5	8.4	7.0
	5%	WB	4.3	4.4	4.1	4.8	7.9	10.4	12.6	13.7	11.5	9.2	6.5	4.6
		MCDB	5.6	5.7	5.9	6.5	10.7	13.1	15.1	16.0	13.3	10.7	7.8	6.2
	10%	WB	3.6	3.8	3.1	4.1	7.2	9.7	11.9	12.9	10.7	8.1	5.2	3.8
		MCDB	4.9	5.0	4.8	5.7	9.1	12.1	13.8	15.2	12.7	9.5	7.0	5.2
Mean Daily Temperature Range	5% DB	MDBR	3.7	3.8	4.4	4.4	4.7	4.7	4.4	4.3	4.2	3.8	3.8	3.8
		MCDBR	3.9	4.4	4.7	5.6	7.9	7.6	6.8	6.3	5.3	4.8	4.8	4.2
		MCWBR	3.3	3.5	3.4	3.8	4.9	4.6	4.4	4.1	3.7	3.7	4.0	3.5
	5% WB	MCDBR	3.5	3.9	4.4	4.7	6.5	6.6	6.0	5.6	5.0	4.4	4.7	3.9
		MCWBR	3.2	3.3	3.3	3.4	4.3	4.2	4.2	3.9	3.8	3.8	4.2	3.4
Clear-Sky Solar Irradiance	taub	0.281	0.344	0.375	0.414	0.417	0.402	0.386	0.383	0.368	0.339	0.327	0.283	
	taud	2.365	2.195	2.117	2.082	2.144	2.293	2.407	2.405	2.377	2.322	2.137	2.271	
	Ebn at Noon	641	695	766	794	819	841	849	826	781	699	519	518	
	Edh at Noon	48	84	116	139	139	122	107	100	90	73	59	41	
All-Sky Solar Radiation	RadAvg	0.45	0.98	2.31	3.44	4.44	4.75	4.27	3.58	2.52	1.38	0.60	0.32	
	RadStd	0.07	0.12	0.37	0.31	0.53	0.49	0.39	0.32	0.23	0.11	0.07	0.03	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page